

# SANA KHAN | Software Developer

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## SUMMARY

Software Developer with 3+ years of experience designing and deploying scalable web, cloud, and mobile applications using Node.js, TypeScript, PostgreSQL, AWS, and Swift. Skilled in microservices, API integrations, CI/CD, DevOps practices, and AI/ML model development and deployment. Experienced in iOS app development, agile collaboration, and building secure, high-performance enterprise solutions that enhance user experience and drive business outcomes.

## SKILLS

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| Programming Languages:     | TypeScript, JavaScript, Python, SQL, HTML, CSS, Swift                                                                                            |
| Frameworks & Libraries:    | Node.js, Express.js, FastAPI, React, REST APIs, GraphQL, Spring Boot                                                                             |
| Databases & Messaging:     | PostgreSQL, MySQL, SQL Server, Redis, Kafka                                                                                                      |
| Cloud & DevOps:            | AWS (EKS, S3), Docker, Kubernetes, Terraform, GitHub Actions, Azure DevOps, CI/CD, Infrastructure as Code (IaC)                                  |
| Monitoring & Security:     | Prometheus, Grafana, ELK Stack, SonarQube, JWT Authentication                                                                                    |
| Collaboration & Processes: | Git/GitHub, Agile/Scrum, SDLC, Unit Testing, Integration Testing, Test Automation                                                                |
| IT Support & Productivity: | Microsoft 365, Device Configuration, Basic Networking, Active Directory, Ticketing Systems (JIRA/ServiceNow), Troubleshooting & End-User Support |
| AI/ML & Data Science:      | TensorFlow, Scikit-learn, Pandas, MLflow, Predictive Modeling, Model Deployment                                                                  |
| Mobile Development:        | iOS Development, Swift, Xcode, UIKit, Core Data, Alamofire, Mobile App Development                                                               |

## EXPERIENCE

### STUDENT IT TECHNICAL SUPPORT | Oklahoma Christian University, OK

Aug 2024 – Aug 2025

- Delivered Tier 1 & Tier 2 technical support to students, faculty, and staff, resolving hardware, software, and account-related issues in a timely manner.
- Assisted with Microsoft 365 applications (Outlook, Teams, OneDrive, and SharePoint) and university infrastructure tools, improving productivity and user adoption.
- Configured, maintained, and troubleshoot end-user devices (laptops, printers, and mobile devices) by performing OS updates, driver installations, and network connectivity checks, ensuring compliance with campus IT standards and minimizing downtime for students and faculty.
- Supported resolution of basic networking and cloud application issues, escalating complex incidents to senior IT staff when necessary.

### SOFTWARE DEVELOPER | Sage Softtech, India

Feb 2023 – Dec 2023

- Developed backend microservices for ERP/CRM/HCM modules using Java, Spring Boot, SQL Server, and Redis, enabling secure and scalable order, inventory, and sales management.
- Built iOS companion app features with Swift, Xcode, Core Data, and Alamofire, supporting offline field-service operations and real-time order capture.
- Integrated AI/ML predictive models for demand forecasting and predictive maintenance using TensorFlow and Scikit-learn, improving inventory planning accuracy.
- Optimized database queries, caching strategies, and implemented RabbitMQ for asynchronous processing, enhancing system throughput and reducing latency under peak loads.
- Automated CI/CD pipelines using Docker and Azure DevOps, deployed microservices to AKS, and configured Azure Application Insights for monitoring, reducing downtime and rollback times.
- Collaborated in Agile teams, performing code reviews, sprint planning, and cross-functional delivery to ensure timely, high-quality releases aligned with business goals.

### IOS Developer Intern | Web Wings Global Solutions Pvt. Ltd, India

Aug 2022 - Jan 2023

- Contributed to the development and maintenance of iOS OTT and live-streaming applications by implementing features in Swift using Xcode and integrating REST APIs for real-time content delivery.
- Assisted in backend integration with APIs and performed functional and regression testing, ensuring seamless playback, secure authentication, and consistent performance across different iOS devices.
- Worked in Agile sprints, collaborating with senior developers and testers to fix bugs, enhance UI/UX features, and document technical workflows for future scalability.
- Gained exposure to cloud-based mobility solutions (AWS-backed services) and cross-team collaboration, strengthening understanding of enterprise-grade app development and deployment practices.

| SOFTWARE DEVELOPER   Addons Softech, India                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Oct 2021 – Jul 2022 |
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| <ul style="list-style-type: none"> <li>Designed and deployed RESTful microservices for e-commerce platforms using Node.js (TypeScript), Express, PostgreSQL, and JWT authentication, improving API response times and enabling secure multi-tenant access.</li> <li>Optimized database queries and caching strategies (PostgreSQL, Redis), reducing page/API latency by 40% and improving throughput during peak loads.</li> <li>Automated CI/CD pipelines with GitHub Actions, Docker, Kubernetes, and Terraform on AWS, cutting release cycles from weeks to days and ensuring zero-downtime deployments.</li> <li>Implemented observability and monitoring with Prometheus, Grafana, and ELK stack, reducing MTTR through proactive alerting and incident tracking.</li> <li>Integrated an AI-powered personalization microservice (Python, FastAPI, Kafka) with recommendation models, boosting user engagement and increasing average order value by 15%.</li> <li>Built ML pipelines for product recommendation and demand forecasting using Scikit-learn and Pandas; containerized and deployed models on AWS for scalable inference.</li> <li>Enforced quality and security standards with unit/integration testing, test automation (Jest, Playwright), Git-based code reviews, and SonarQube checks, improving code coverage and minimizing production regressions.</li> <li>Collaborated in Agile/Scrum teams, participating in sprint planning, backlog grooming, and cross-functional delivery, ensuring timely releases aligned with business goals.</li> </ul> |                     |

EDUCATION

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| Master of Science in Computer Science, Oklahoma Christian University           | Aug 2025 |
| Bachelor of Technology in Computer Science, Koneru Lakshmaiah University (KLU) | Apr 2023 |

PROJECTS

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| <b>KL Atlas – Interactive University Map</b> <ul style="list-style-type: none"> <li>Developed a web app with ML-based route optimization for campus navigation using JavaScript and HTTP protocols.</li> <li>Optimized data preprocessing and performance tuning for smooth map interactions.</li> </ul>                                                                        |  |
| <b>Agricultural Genie – Smart Farming Assistant</b> <ul style="list-style-type: none"> <li>Built a Python Flask platform integrating ML models for crop selection and pesticide recommendations.</li> <li>Deployed solution on local cloud with API integration for real-time insights.</li> </ul>                                                                              |  |
| <b>LLM Classification Fine-Tuning – Kaggle Competition</b> <ul style="list-style-type: none"> <li>Fine-tuned BERT model with Hugging Face Transformers for text prompt classification.</li> <li>Implemented end-to-end NLP pipeline with tokenization, training, and transfer learning.</li> </ul>                                                                              |  |
| <b>Thai Speech Recognition &amp; Translation</b> <ul style="list-style-type: none"> <li>Built an ASR + MT pipeline using Wav2Vec2 (speech-to-text) and MarianMT (translation) for Thai → English.               <ul style="list-style-type: none"> <li>Deployed real-time transcription &amp; translation via Gradio web app with preprocessed datasets.</li> </ul> </li> </ul> |  |

CERTIFICATIONS

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| <ul style="list-style-type: none"> <li>Certified Solutions Architect – Associate</li> <li>Oracle Cloud Infrastructure Developer Associate</li> <li>Automation Anywhere Certified Essentials RPA Professional (AA-Essentials &amp; AA-Advanced)</li> <li>Google Certified TensorFlow Developer</li> <li>Artificial Intelligence Foundations: Neural Networks - LinkedIn (Issued Oct 2021)</li> <li>Designing for Neural Networks and AI Interfaces - LinkedIn (Issued Oct 2021)</li> <li>Learning Spring with Spring Boot - LinkedIn (Issued Oct 2021)</li> <li>Programming Foundations: Fuzzy Logic - LinkedIn (Issued Oct 2021)</li> <li>Training Neural Networks in Python - LinkedIn (Issued Oct 2021)</li> <li>Java EE: Java Server Faces JSF - LinkedIn (Issued Apr 2021)</li> <li>Learning Java Enterprise Edition - LinkedIn (Issued Apr 2021)</li> <li>Become a Django Developer - LinkedIn (Issued Mar 2021)</li> <li>Building RESTful Web APIs with Django - LinkedIn (Issued Mar 2021)</li> <li>Building a Personal Portfolio with Django - LinkedIn (Issued Mar 2021)</li> <li>Django: Forms - LinkedIn (Issued Mar 2021)</li> <li>Learning Django - LinkedIn (Issued Mar 2021)</li> <li>Test-Driven Development in Django - LinkedIn (Issued Mar 2021)</li> </ul> |  |
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